

Self-Improving Research Agent

A research agent that *learns* from its own mistakes.

Ask a question. It plans, researches with real tools, writes a cited report, checks every number, gets reviewed by a critic, and saves lessons for the next run.

THE PROBLEM · REAL FAILURES FROM EARLY RUNS

1 Wrong source
Price copied from a blog when the task said "official page"

2 Mental arithmetic
 $10 \times \$0.33$ written as \$3.33

3 Misattributed citation
A number cited to a source that doesn't contain it

4 Unit mix-up
"per 1M dimensions" shown as "per GB-month"



RUN #1 · FIRST DRAFT

10 GB $\times$ \$0.33/GB = **\$3.33** per month [1][2][3]

CITATION CHECK · NO LLM

FAIL not_in_source: \$3.33 not found in any cited source, and not computed with the calculator

CRITIC · GPT-OSS-120B

3/10 · needs_research

"the task required the official pricing page"

THE GOAL

An agent that checks its own work — and doesn't repeat the same mistake twice.

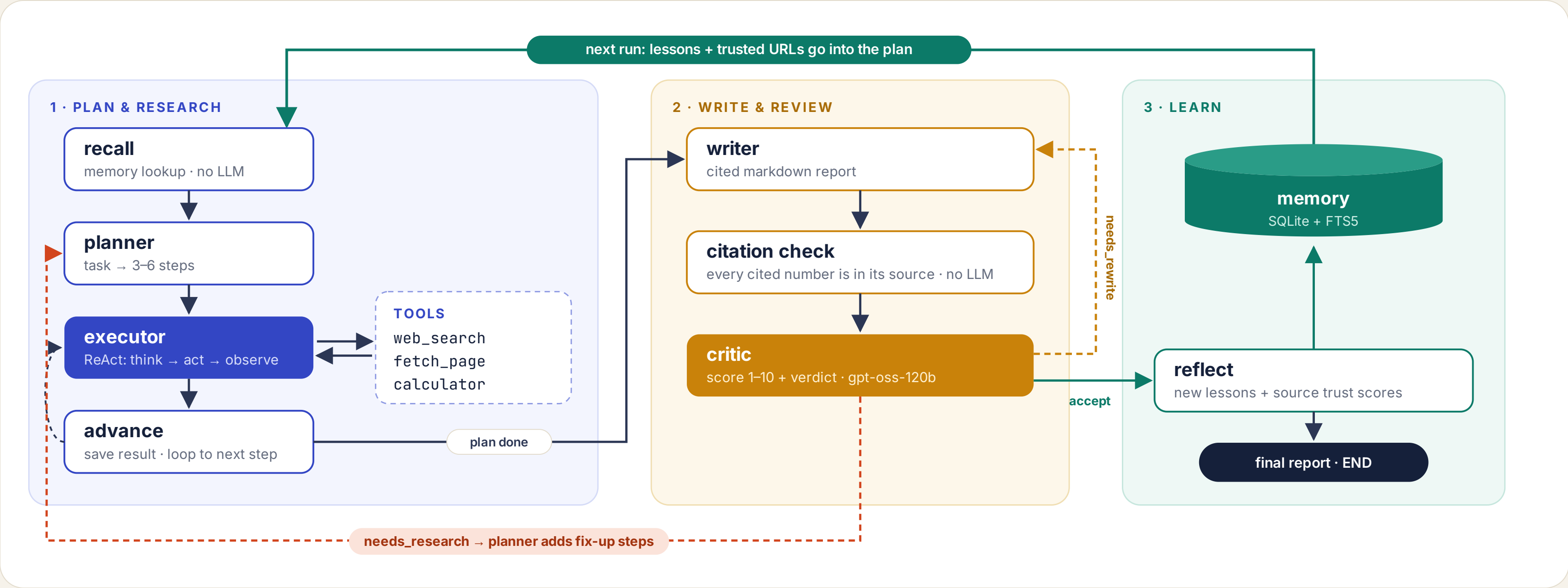


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Plan → research → write → review → *learn*

One shared state across every node. The critic routes to more research, a rewrite, or done.



Plan-and-Execute planner

ReAct loop executor

Tool use search · fetch · calc

Critic + smart routing ≤ 2 revisions

Reflection + long-term memory reflect · recall

The fourth run got it right *the first time*.

3 → 9

first critique score, out of 10

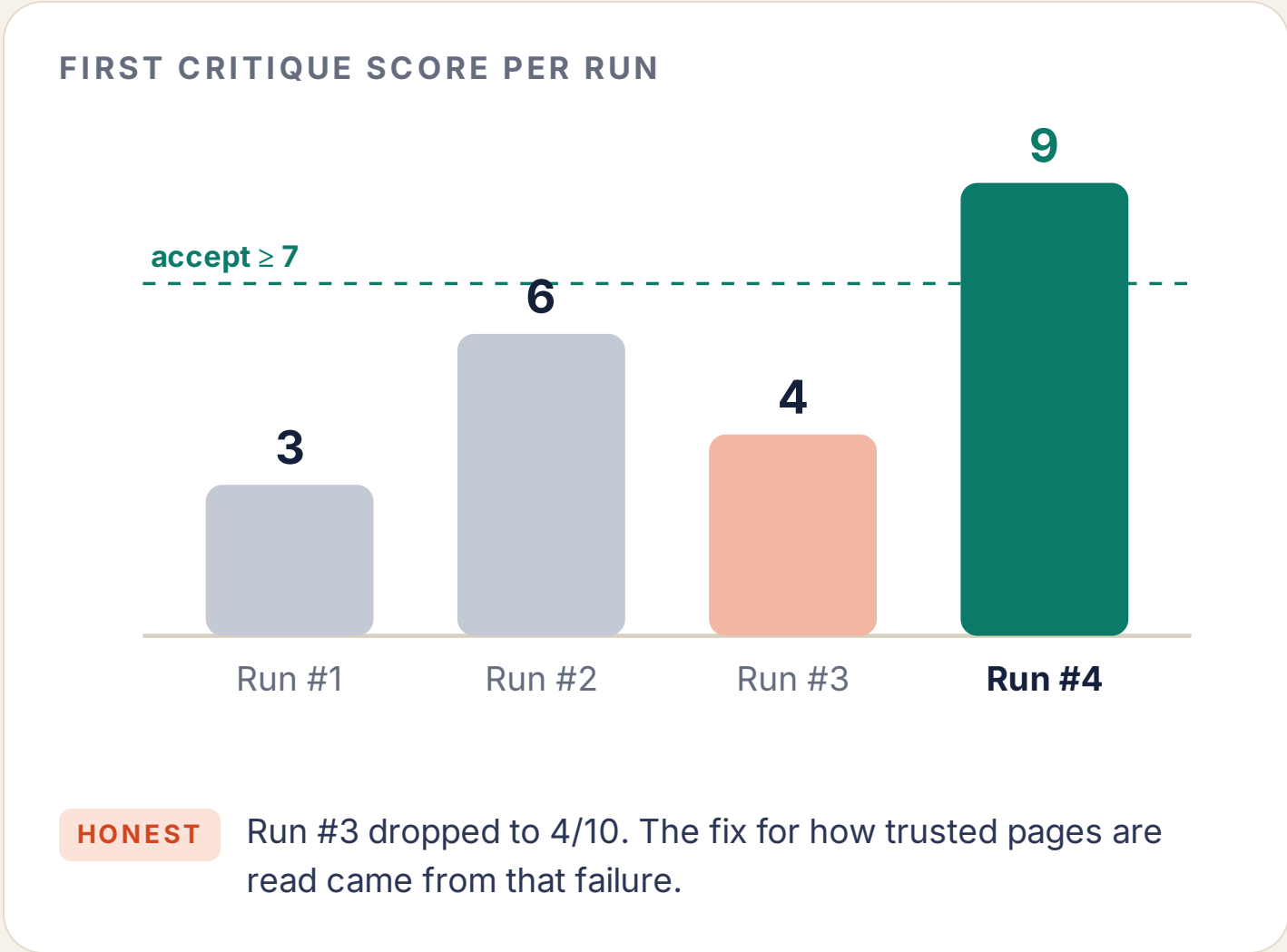
1 → 0

revisions needed

-52%

tokens used · 28,411 → 13,604

	RUN #1 · EMPTY MEMORY	RUN #4 · WITH MEMORY
First critique	3/10 needs_research	9/10 accept
Revisions	1	0
Tokens	28,411	13,604
Memory recalled	nothing	4 lessons + 1 trusted URL
Plan step 1	web search	fetch the official pricing page
SETUP Both runs: qwen3.8-27b for every role, critic gpt-oss-120b, fresh memory database, model fallbacks off. Only the memory changed.		



What you see *in the app*

Live at asad-research-agent.streamlit.app — run a task, or replay a recorded one.



Live plan checklist

Steps tick off as they finish.
Steps the critic asked for are marked `fix-up`.

RESEARCH TAB



Step-by-step trace

memory · plan · think · act ·
observe · verify · critique · learn
— as they happen.

RESEARCH TAB



Score + cited report

Critic score, revisions, the report
with a download button, and
tokens per model.

RESEARCH TAB



Memory tab

Runs chart, lessons with used /
helpful counts, and trusted
sources with verified facts.

MEMORY TAB



Replay mode

Any recorded run, step by step,
with zero API calls — a demo that
survives quota limits.

SIDEBAR · REPLAY



Same story in the CLI

```
python cli.py "task" prints  
PLAN, THINK, ACT, OBSERVE,  
VERIFY, CRITIQUE, LEARN.
```

TERMINAL · CLI.PY

6/6

known failure cases caught by the citation check + critic

- ✓ Unit error: per 1M dimensions shown as
per GB
- ✓ Contradiction: Qdrant "\$25 minimum" vs
"no minimum"
- ✓ Contradiction: Pinecone \$16/M vs
\$0.00000025/RU
- ✓ Misattribution: \$0.33 cited to a source
without it
- ✓ Missed constraint: official page never
fetched → fix-up step
- ✓ No false alarm: calculator result \$3.30 not
flagged

+ clean control report accepted (8/10)

Built entirely on *free tiers* — and what that costs

■ Stack

ORCHESTRATION

Python 3.11

LangGraph

MODELS

Groq gpt-oss-20b / 120b

qwen3.8-27b

NVIDIA NIM

Gemini 2.5 Flash

TOOLS & DATA

Tavily

DuckDuckGo

SQLite + FTS5

UI & DEPLOY

Streamlit Cloud

■ Resilience

- 5-model fallback chain across separate free quotas
- Rate-limit retry using the provider's own wait time
- Graceful stop with a partial report — never a crash
- Public demo capped at 15 live runs/day · 81 unit tests

■ Limits

- **Quotas:** ≈200k tokens/day per Groq model; a broad comparison costs 70–100k
- **Small models:** gpt-oss-20b rarely fetches pages without memory
- **Evidence:** one task family, four runs — a demonstration, not a benchmark
- **Recall:** keyword-based (FTS5), not semantic

■ Next

- Semantic recall with embeddings
- Prune lessons that don't help; re-check trusted sources on a schedule

Thank you.

LIVE asad-research-agent.streamlit.app

CODE github.com/shaikasadahmed2k23/self-improving-research-agent_Submission

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